R16

Code No:131AK

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD
B.Tech I Year I Semester Examinations, December - 2018
BASIC ELECTRICAL AND ELECTRONICS ENGINEERING
(Common to EEE, ECE, CSE, EIE, IT, ETM)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

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PART - A

(25 Marks)

1.a) For the circuit shown in figure 1. Find the relation between v_1 and v_2. [2]

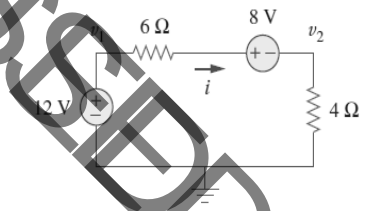


Figure: 1

b) Explain Source transformation with the help of neat diagrams. [3]
c) Write the statement of Super Position Theorem. [2]
d) What is Q-Factor and explain its significance? [3]
e) How does a diode appear as in its forward region of its characteristic? Explain. [2]
f) How does junction capacitance of a linearly graded junction varies with the applied reverse voltage V_R? [3]
g) Differentiate between the terms h_{FE} and h_{fe}. Write the relationship between them. [2]
h) Draw the h-parameters model of transistor. [3]
i) Give different regions of operation of JFET. [2]
j) What is pinch-off voltage? Give its equation. [3]

PART - B

(50 Marks)

2.a) Find V_o in the circuit, shown in figure 2.

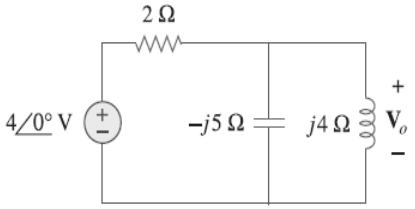


Figure: 2

b) Define and give symbols of the following:
i) Dependent sources
ii) Independent Sources
iii) Practical Sources
iv) Ideal Souces [6+4]

OR

3.a) If Req=50 ohm, in the circuit shown in figure 3 find R?

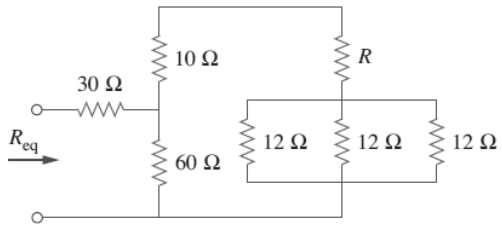
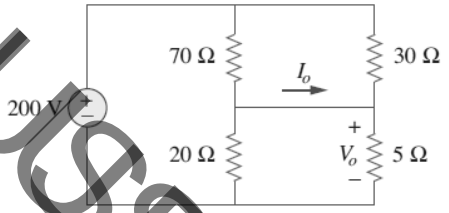


Figure: 3

b) Calculate V_o and I_o in the circuit shown in figure 4.

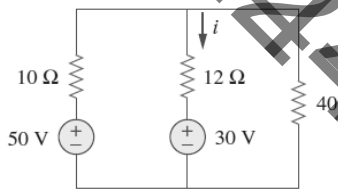


[5+5]

Figure: 4

4.a) A parallel Resonant circuit has R= 5KΩ, L=8mH, and C= 60μF. Determine:
i) Resonant frequency
ii) The band width
iii) The Quality Factor

b) Solve for '*i*' using Norton's Theorem in the circuit shown in figure 5.



[5+5]

Figure: 5

OR

5.a) Find v_o using super position theorem in the circuit shown in figure 6.

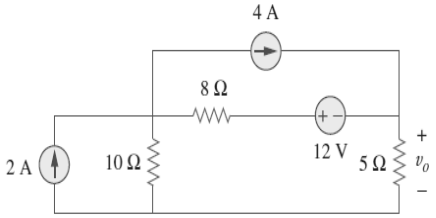
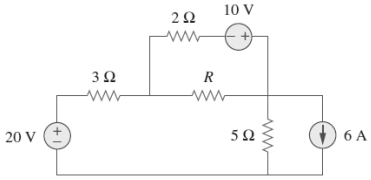


Figure: 6

b) Find the Maximum power that can be delivered to R in the circuit shown in figure 7.



[5+5]

Figure: 7

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6.a) Compare C, L, L-Section, Π-Section (CLC and CRC) Filters in all respects.
b) Explain the operation of Half wave Rectifier with the help of neat diagrams. [6+4]

OR

7.a) Obtain different equivalent Circuits of a PN Junction diode.
b) Explain the VI characteristics of PN Junction diode with neat diagrams and explain. What is Static Resistance and Dynamic Resistance? [5+5]

8. Draw CE Configuration with Voltage – divider bias and then analyze 0^0t of Q - point and stability factors. [10]

OR

9. Illustrate the input and output characteristics of BJT in three configurations. [10]

10.a) Compare BJT and JFET devices in all respects.
b) Obtain the expression for the pinch off voltage V_p in case of n-channel JFET. [5+5]

OR

11. Describe the principle of operation and V-I characteristics with the help of energy band diagrams. Highlight negative resistance region on the V-I characteristics. [10]

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